Marçal Herraiz Bayó

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Education

MSc in Mathematics – PSL University — Paris, France
Rank: 5/116 (top 5%); GPA: 15.052/20; Minor in Physics — Sept. 2025 – Present

- Coursework at Dauphine and École Normale Supérieure. Full FSMP scholarship.
- First-year thesis: "Market Making and Optimal Inventory Management" (18/20).

BSc in Mathematics – University of Barcelona — Barcelona, Spain
Rank: 5/141 (top 4%); GPA: 9.02/10 — Sept. 2020 – June 2025

- Thesis: "The Riemann–Hilbert Correspondence for Flat Connections on Principal Bundles" (10/10).

BSc in Physics – University of Barcelona — Barcelona, Spain
Rank: 16/196 (top 8%); GPA: 8.79/10 — Sept. 2020 – June 2025

- Thesis: "Improving Classical Shadows with Grouping Strategies" (9.7/10).

Research and Industry Experience

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
Incoming Visiting Student – Particle Physics — Apr. 2027 – Oct. 2027

- Will study Higgs boson decays into two muons using data from the CMS detector at CERN.
  Supervisor: Prof. Dr. Christoph Paus.

Westinghouse Nuclear — Madrid, Spain
Simulation Modeling Engineer – Nuclear Plant Simulation — July 2025 – Present

- Developing a compressible liquid–vapor–air model in Fortran with revised state variables and pressure solver.
- Secured leadership approval for a new multiphase-modeling project by presenting preliminary results.
- Built and validated a condenser heat-transfer model with under 2% error against plant data.
- Resolved major energy-conservation errors in open tanks by implementing a new overflow model.
- Delivered thermal-hydraulic models to production software for projects and plants under construction.

Barcelona Supercomputing Center (BSC) — Barcelona, Spain
Visiting Student – Quantum Computing — Nov. 2024 – June 2025

- Implemented a hybrid algorithm combining classical shadows with observable grouping using PennyLane.
- Improved energy-estimation accuracy up to 18-fold over standard classical shadows.
- Reduced measurement requirements by a factor of 100–1,000 compared with standard classical shadows.
  Supervisors: Prof. Dr. Bruno Julià, Dr. Berta Casas, Dr. Sergi Masot.

Institute for Research in Biomedicine (IRB) — Barcelona, Spain
Research Intern – Computational Biophysics — July 2024 – Feb. 2025

- Modeled DNA flexibility using coarse-grained computational methods in C.
- Extended the model to analyze how DNA damage affects flexibility.
- Developed Python algorithms to compute DNA geometric descriptors.
  Supervisor: Prof. Dr. Modesto Orozco.

Center for Mathematical Research (CRM) — Barcelona, Spain
Research Intern – Complex Dynamics — June – Aug. 2023

- Studied bifurcations in the complex exponential family and the non-structural stability of $\exp(z)$.
  Supervisors: Prof. Dr. Núria Fagella, Prof. Dr. Xavier Jarque.

Honors and Awards

Fondation Sciences Mathématiques de Paris: full master's scholarship; 1,000+ applicants. — 2025
Excellent with Honors: awarded in 16 subjects across both BSc programs at UB. — 2021–2025
Collaboration Scholarship: research toward the Mathematics BSc thesis. — 2024
Maths4Life Award: one of five IRB grants for research in mathematics and biomedicine. — 2024
Honors in Baccalaureate: academic distinction covering first-year university tuition in full. — 2020
Mad for Science Programme: selective Physics research programme for high-school students. — 2019

Skills

Programming	Proficient: Python (NumPy, SciPy, pandas, scikit-learn), Fortran, C; familiar: R, Bash.
Numerics	Monte Carlo & variance reduction, ODE/SDE solvers, finite differences, particle methods.
Inference & ML	Regression, regularization, Bayesian inference, PCA, SVMs, NNs, graphical models, time series.
Development	Debugging, model validation, legacy-code refactoring, code review, technical documentation.
Software	Git/GitHub, Unix/Linux, Jupyter, PennyLane, LaTeX, Mathematica, SageMath.
Languages	Spanish and Catalan (native) • English (C1) • French (A2).